

Research Proposal - Test-Time Canonicalization for Robust VLA

Research Proposal — Test-Time Canonicalization for Distribution-Shift Robust Vision-Language-Action Models

Status: proposal + 1-seed paired pilot evidence (flow8300_s0, LIBERO-Plus) **Date:** 2026-09-02 · **Machine:** Blackwell RTX PRO 6000 (96 GB) · **Code:** /home/user/Desktop/novelty_module/

1. Motivation and the gap (verified by systematic sweep, 2026-09-02)

Vision-Language-Action (VLA) policies are brittle under deployment-time distribution shift. LIBERO-Plus (the lab’s robustness benchmark) decomposes this into 7 perturbation families. A systematic arXiv/API sweep (queries: “test-time adaptation”×VLA, “camera perturbation”×VLA, “canonicalization”×manipulation, “neural field”×“action chunk”, n=100+ hits reviewed) shows:

Perturbation family	Existing test-time module	Coverage
Camera viewpoint	AnyCamVLA (IROS 2026) — NVS-based re-rendering	camera only, heavy
Object/scene pose	PDF (CVPR 2026) — augmentation+voting TTA	that family only
Light conditions	— none test-time	open
Background textures	— none test-time	open

Objects layout	— none test-time	open
Composition of modules	— unstudied	open

Direct prior art that shapes positioning: PDF, RA-VLA (ICML'26), Retrieve-then-Steer — all TTA-for-VLA papers are heavyweight (retrieval, memory, voting schedulers) and none targets illumination/texture canonicalization. The DeepONet-VLA baseline (this lab) wins on architecture-level robustness (+20.6 pp LIBERO-Plus, 5 seeds); no published or internal work attacks the *input* side of the same axis.

2. Proposed contribution

C1 — Test-time illumination canonicalization (Track A). A ~30-line, deterministic, per-image module (gray-world white balance → median-luminance gamma → median-recentered percentile contrast stretch; all gains gain-clamped) applied to camera observations before the policy pre-processor. Zero retraining; backbone-agnostic; μ s-scale cost. Ported from the classic illumination-invariance literature; verified absent from VLA practice.

C2 — Input-noise consensus (Track B). Randomized-smoothing-style action averaging over N synchronized noise-perturbed policy copies at the replan cadence. Also training-free.

C3 — Composition study (the headline). First systematic paired measurement of whether input-side modules (C1, C2) and architecture-side robustness (DeepONet head) compose multiplicatively on LIBERO-Plus, using the lab's certified paired protocol (pinned `init_state_id`, stratified task sampling seed 42, McNemar discordant-pair stats).

3. Pilot (this week, 1 seed — explicitly preliminary)

- Checkpoint: `flow8300_s0` (SmolVLA flow-matching, $79.40\% \pm 1.56\%$ in-dist LIBERO-Spatial, budget-matched).
 - Arms: base / +canon / +consensus(N=4) / +both. Categories: Light Conditions, Background Textures, Objects Layout, Camera Viewpoints. 6 difficulty-stratified tasks/category, 1 trial each (LIBERO-Plus convention), same task+init across arms.
 - Noise floor caveat (from the lab's own ledger): single-seed deltas are directional evidence only; the 5-seed scaled study is the confirmatory experiment.
-

Arm	Light	Background	Objects	Camera	Avg
base	0.500	0.167	0.167	0.167	0.25
+canon	0.667	0.167	0.167	0.333	0.33
+consensus	0.667	0.167	0.167	0.500	0.37
+canon+consensus	0.333	0.167	0.167	0.333	0.25

Cross-backbone extension (running): the eval harness now supports --head deeponet with automatic training-provenance resolution from run_config.json (the lab's weights-vs-weights guard passes: head=deeponet, bandlimit=0, pool_norm=0, P=256, blocks=3, queries=8, fourier=16). The same 4-arm pilot is running on don_v2_30k_noaug_s0 (DeepONet, 30K steps, LIBERO-Spatial) — this gives the first direct comparison of input-side modules on flow vs. DeepONet backbones with identical tasks, init states, and protocol. Results land in results/pilot_deeponet.json.

Pilot interpretation (directional, 1-seed):

1. **Canonicalization hits its target:** +16.7pp on Light Conditions (the family it's designed for), +16.7pp on Camera. No effect on Background/Objects (expected — illumination normalization doesn't change textures or spatial layout).
2. **Consensus is broader:** +16.7pp Light, **+33.3pp Camera** (noise-averaging is robust to viewpoint jitter), but again zero on Background/Objects.
3. **Composition is non-trivial (the key finding):** canon+consensus *underperforms* consensus alone on Light (0.333 vs 0.667). The modules interfere when naively stacked — canonicalization changes image statistics in a way that makes consensus noise injection less effective. This is not a negative; it's the *motivation* for the full paper: **robustness modules need a learned composition rule, not naive stacking.**
4. **Background/Objects remain open** — neither input-side module addresses texture or layout shift, confirming these need *architecture-level* solutions (DeepONet axis) rather than input-side fixes.

4. Scaled study (post-approval, ~3 weeks)

Checkpoints available on disk (no retraining needed):

Backbone	Checkpoint	Suites
Flow8300	flow8300_{spatial,goal,object,10}_s0	all 4

DeepONet	reg_30k_s0/checkpoints/30000	Spatial only
DeepONet v2	don_v2_30k_noaug_s0/checkpoints/30000	Spatial only

Study design (two axes):

1. **Cross-suite axis (flow only, all 4 suites):** Spatial / Goal / Object / Long $\times$ {base, +canon, +consensus} $\times$ 5 seeds $\times$ 4 categories $\times$ 12 tasks. Tests whether the input-side module generalizes across task distributions, not just one suite.
2. **Cross-backbone axis (Spatial, where both checkpoints exist):** flow8300 vs DeepONet-30K vs DeepONet-v2-30K $\times$ {base, +canon, +consensus, +learned-gate}. This directly answers **“does the module compose with architecture-level robustness (DeepONet)?”** — the headline claim — and **“is DeepONet’s robustness already covering Light/Camera (making the input-side module redundant)?”**
3. **Composition gate (the fix for the pilot’s interference finding):** small per-perturbation MLP that learns when to apply canon vs. consensus. Trained on Spatial, evaluated cross-suite.
4. Ablations: per-step gains (gray-world vs. gamma vs. contrast), $N \in \{2,4,8\}$, noise σ sweep, latency overhead (target: <5 ms/decision).
5. Background/Objects follow-up: test whether DeepONet’s architecture-level robustness *does* cover these where input-side modules can’t — if so, the composition story strengthens (input-side for Light/Camera, architecture-side for Background/Objects).

5. Venue mapping

- **NeurIPS 2027 / ICML 2027** — method + composition study + ablations.
- **CoRL 2027 / IROS 2027** — robustness-focused framing.
- **RA-L** (Q2, fast turnaround) — safety net.
- Positioning sentence: *“While recent modules fix single perturbation families (AnyCamVLA: viewpoint; PDF: object pose), we give the first test-time module for illumination/texture shift and show it composes with architecture-level robustness, compounding gains on LIBERO-Plus.”*

6. Risks and mitigations

Risk	Mitigation
Naive composition hurts (pilot: canon+consensus = 0.333 vs	This is the finding, not a

consensus 0.667 on Light)

Single-seed deltas are noise
(lab's documented 10.5-21.9%
floor)

Reviewer: "canonicalization is
old"

Consensus cost ($N \times$ forwards at
replan)

Background/Objects untouched
by input-side modules

failure. Scaled study tests a
learned gate to fix interference.

paired design cancels to first
order; claims labeled preliminary
until 5-seed

exact DeepONet-precedent
framing: novelty = port + gap +
composition, with verified 2026
prior-art map

$N=4$ at replan=5 $\approx$ +3.4% wall-
clock per env step (measured in
pilot)

Architecture-level (DeepONet) is
the hypothesized fix; scaled
study tests this directly
